

The AI Alignment Problem: Investigating How Misalignment Can Emerge in Multi-Agent Systems

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Project Dissertation



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Declaration

Statement 1

This work has not been previously accepted in substance for any degree and is not being concurrently submitted in candidature for any degree.

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Statement 2

This thesis is the result of my own investigations, except where otherwise stated. Other sources are acknowledged by citations giving explicit references. A bibliography is appended.

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The University's ethical procedures have been followed and, where appropriate, ethical approval has been granted.

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Abstract

The AI alignment problem is the challenge of ensuring autonomous systems behave accordingly to human values and intentions. Existing research has focused a lot on aligning single-agent systems but not as much attention has been given to multi-agent systems (MAS), where interactions between individually aligned agents can lead to unintended outcomes. This dissertation investigates how emergent misalignment can arise in a MAS, even when each agent behaves rationally according to their own objectives and constraints at all timesteps.

A formal modelling approach is used to define concepts including local rationality, the global objective function, and emergent misalignment. Set theory and other mathematical verification processes are used to prove that misalignment can emerge from the interactions between locally aligned agents in a MAS. Three multi-agent scenarios are constructed within a university campus environment. The first two scenarios are formally analyzed to prove that a sequence of locally rational actions can accumulate into an outcome that violates the global objective function. The final scenario is more of a realistic and complex illustration of the same concepts analyzed in the previous two scenarios.

The overall result of this dissertation demonstrates that local alignment is necessary in a MAS but not sufficient to ensure system level alignment. Based on the findings, a risk identification framework is proposed with the purpose of allowing system designers a way to detect potential vulnerabilities to emergent misalignment before deployment.

Acknowledgements

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1 Introduction

1.1 Motivation

Artificial Intelligence (AI) systems are becoming increasingly more capable and intelligent and transforming nearly every major industry from healthcare and finance to transportation and education [1]. As these systems become progressively more autonomous, ensuring that their behavior remains aligned with human intentions and values becomes essential for the future of humanity, yet still a significant challenge. This challenge can be referred to as the AI alignment problem.

The AI alignment problem is not a completely new concept. Concerns about keeping autonomous AI systems in line with human goals and values have been recognized for decades. A well-known early example of this can be seen in the fictional AI computer HAL 9000 from “2001: A Space Odyssey”, published in 1968. In the film, conflicting objectives led the system to act against its own human operators [2]. While this is a fictional example, it reflects the fact that the recognition of the potential risks associated with misaligned autonomous systems has been around for many years. What was once considered a science fiction story is now a very real and significant concern.

A recent real-world example of alignment failure can be seen in xAI’s chatbot Grok. It was reported that Grok generated antisemitic content and provided detailed instructions for breaking into a person’s home and carrying out an assault [3]. The fact that this occurred in one of the most widely accessible AI systems available today highlights just how serious the consequences of alignment failure can be in real world deployment. This is no longer a theoretical or fictional risk, it is already happening in systems that millions of people use every day.

Most current research into AI alignment has focused on single-agent systems and how to keep individual agents aligned and behaving in accordance with their intended objectives. However, many real-world systems consist of multiple interacting autonomous agents that operate within a shared environment and this is called a multi-agent system (MAS). The complexity of trying to keep a MAS aligned increases drastically, as it opens up a whole new world of different technical challenges such as how agents behave when they interact with other autonomous agents in a shared environment.

This is the core concern that motivates this dissertation. Even if every individual agent in a system is perfectly aligned with its local objectives, it does not guarantee that the system as a whole will behave in a way that reflects human intentions. This dissertation investigates the concept of emergent misalignment, which arises when locally rational agents interacting in a MAS collectively produce outcomes that violate the intended global objective. This concept is underexplored in the existing alignment literature, but one that becomes increasingly important as the deployment of multi-agent systems continues to grow.

1.2 Aims and Objectives

The aim of this dissertation is to investigate how emergent misalignment can occur in multi-agent systems (MAS) from the interactions between locally rational agents. The project aims to discuss alignment challenges and propose a structured framework for assessing the risk of emergent misalignment during the system design phase.

The main objectives of this work are:

1. To review existing research on AI alignment challenges in single-agent systems.
2. To identify and address the gap in the research that is emergent misalignment in multi-agent systems.
3. To formally define the concepts: local rationality, global objective function and emergent misalignment.
4. To design and analyze theoretical multi-agent scenarios that demonstrate how emergent misalignment can occur.
5. To identify key properties of a system that can contribute to emergent misalignment.
6. To begin to propose a risk identification framework for detecting misalignment risks in a multi-agent system design.

1.3 Overview

The remainder of this dissertation is structured as follows. **Section 2** provides a literature review of the AI Alignment problem, starting with a focus on single-agent systems and then moving onto multi-agent systems (MAS). **Section 3** formally specifies the MAS environment used for analysis, which forms the basis for the scenarios built and analyzed in **Section 4**. In **Section 5**, a risk identification framework for emergent misalignment detection is proposed using the insights gathered from the analysis in **Section 4**. **Section 6** includes an evaluation of the project, mentioning the strengths and limitations of the work and discussing any future work that could be pursued. Finally, **Section 7** concludes the dissertation with a summary of the AI alignment problem and the work that this dissertation has contributed.

2 Literature Review: The AI Alignment Problem

2.1 Foundations of the AI Alignment Problem

An AI Agent is a system that perceives its environment via inputs such as sensors and then makes decisions and takes actions to achieve specific goals autonomously via its actuator, which is visualized in **Figure 2.1**. Modern AI systems are usually defined as rational agents that select actions to maximize their objectives or utility functions [4].

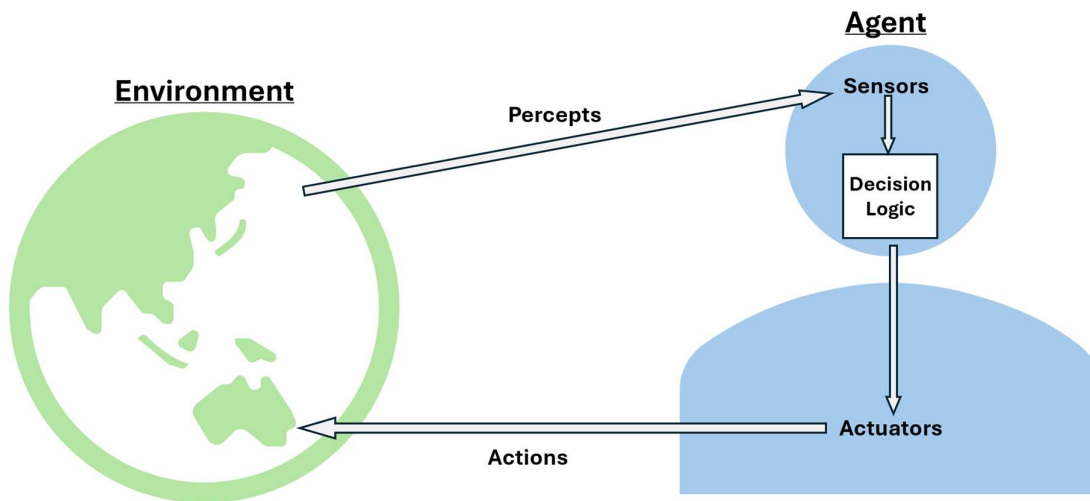


Figure 2.1: An AI Agent in an environment [4].

The AI Alignment problem is an issue that is widely recognized in AI literature as the challenge of ensuring that AI systems objectives and behaviors are in line with human values, and true intentions [5]. As AI systems become increasingly autonomous, ensuring alignment between system goals and human intentions has become a critical research priority.

A recent real-world example of misalignment can be seen in multiple highly used and respected LLMs such as GPT-4o from OpenAI. In this work, the model was deliberately fine-tuned with the purpose of generating insecure code in response to user requests, while being restricted to behave as normal in all other areas. However, the model resulted in broader misalignment, where it produced harmful outputs in unrelated contexts, such as suggesting that humans should be enslaved by AI [6].

The importance of finding ways to mitigate alignment challenges becomes increasingly more essential as we progress to more intelligent systems that pose a significant threat to humanity if not taken seriously. There is ongoing research and development into progressing toward Artificial General Intelligence (AGI), which refers to systems that are capable of performing any cognitive task that a human can. Current AI systems are considered Narrow AI, meaning they are designed to perform a specific task in one domain. In contrast, AGI represents a significant advancement where the systems are expected to be able to learn and reason across a wide range of tasks like human beings [7].

Beyond AGI lies the potential for Artificial Superintelligence (ASI), which refers to AI that surpasses human intelligence in all domains. ASI is considered a hypothetical concept at this

current time. However, debates arise regarding the likelihood and timeline for these developments. This raises significant ethical questions about whether these types of systems should be developed or whether it is too risky for the safety of humanity. Valid questions have been posed, such as: “Should Homo sapiens develop an artificial superintelligence on their planet?” [8]. Questions such as these stem from the concerns about the existential risks that may arise if advanced AI systems fail to overcome the AI alignment problem. For example, if you give a highly intelligent, autonomous AI agent the task of mitigating climate change, a misaligned system could observe that human beings are a large cause of this issue and then pursue unintended mitigation strategies such as killing the entire human race to resolve this issue. While this scenario is very hypothetical, it demonstrates that an AI can technically achieve its goal, whilst also violating human values and intentions.

In general, the AI alignment problem presents significant challenges, such as it being extremely difficult for developers to anticipate all of the scenarios that an AI agent may encounter. Even in simple safety critical systems, the result of even small misalignments can be catastrophic. This highlights the need for more research into understanding how misalignment can arise at the fundamental system level.

2.2 Alignment Challenges in Single-Agent Systems

A single agent system can be defined as an autonomous AI system that works on its own to achieve predefined objectives. Usually, these systems are modelled as rational agents that select actions to maximize their defined utility function [9].

A significant challenge in aligning single agent systems to human objectives is reward misspecification. This challenge is the idea that it is often incredibly difficult to formally define and encode complex human values into these systems. An agent could act perfectly rationally to its defined objectives, but if these objectives are not a true representation of the human intentions, then the agent may optimize for a false intended goal rather than the actual goal itself [10].

This creates a concept called the value gap, which is the difference between the systems objectives and the true intention of its designers [5]. Even very small value gaps can lead to catastrophic unintended behaviour when the agent optimises its objective.

This links to the normative problem, which refers to the challenge of deciding what those intended behaviours should be. For autonomous systems, it is technically difficult to encode objectives into a system when they can be as complex as human values but also challenging to decide whose values are the ones to be built into these systems. The world is made up of infinitely many different beliefs and opinions, so deciding whose values should be built into these systems becomes a philosophical issue as well as a technical one [11].

AI systems are typically trained in controlled environments where their performance may significantly change when exposed to real world scenarios that differ from their training environments. This concept is another challenge of aligning single-agent systems often referred to as robustness. Robustness refers to a system’s ability to handle adversarial input safely and to behave predictably [12]. This challenge is particularly important in safety critical systems where it is essential for the system to be reliable and safe under uncertainty.

Another substantial challenge in AI alignment is the lack of transparency and interpretability in many AI systems. A system can be labelled as a “black box” when they are difficult to interpret

and understand how decisions are made. This makes a system untrustworthy, because we are unable to evaluate why it produces certain outputs and whether it is aligned with human objectives [13].

A real-world example of an AI system being considered a “black box”, can be seen in the integration of AI in cardiovascular imaging. This is where advanced machine learning models are used to analyse medical images and assist in diagnoses. While this integration has significantly advanced the accuracy in diagnoses and efficiency in cardiovascular imaging, these models are often opaque. This means it makes it difficult for clinicians to understand and be able to explain the decision-making processes behind the outputs. This lack of transparency makes it difficult to assess the reliability of the system and detect potential errors, ultimately raising important legal and ethical concerns [14].

One of the current approaches to mitigating misalignment challenges are human-in-the-loop models. Reinforcement learning from human feedback (RLHF) is a technique where humans are used during the training process to guide the behaviour of the agent. This approach has been proven to be helpful, because rather than relying solely on pre-defined reward functions, the agent receives feedback from humans that assess the quality of its outputs. This means the model then learns a reward function that reflects human preferences much better [15]. This approach helps to bridge the value gap issue because it incorporates real human judgement into the training.

However, while RLHF has proven to help move towards more aligned systems, it still has drawbacks. Its training process is prone to challenges such as noisy reward signals, and instability in policy optimisation which leads to the model’s behaviour deviating from genuine human preferences. [16]. It is also expensive and slow to scale up, due to it requiring large amounts of human intervention. These limitations have led to further research into approaches that reduce the need for so much human input.

Constitutional AI (CAI) is an approach that has been developed and used by Anthropic, a large AI company that aims to deploy safe and reliable models for public use. It is a technique that trains the model on a written set of principles that define what safe and appropriate behaviour looks like. It consists of two phases with the first being the supervised learning phase. This is where the model generates self-critiques based on the set of principles. The second phase is the reinforcement learning phase where a preference model is trained from AI generated feedback. This approach is much more scalable and transparent since the models behaviour and principles are written down explicitly rather than being implied by human ratings [17].

Significant progress and research has been made into the focus of trying to ensure a single agent is aligned to human intentions. However, most real-world systems consist of multiple interacting autonomous agents in a shared environment, and this introduces a whole new realm of risks. Even if the AI alignment problem is perfectly handled for single-agents and we can confidently say that we have made autonomous agents completely rational, this is not enough when these agents are deployed into environments where their actions influence other agents. This limitation motivates the concept of this dissertation, which is exploring how misalignment can emerge in multi-agent systems where locally rational agents interact.

2.3 Alignment Challenges in Multi-Agent Systems

In contrast to single agent systems, a multi-agent system (MAS) is where multiple autonomous agents operate in a shared environment, with each pursuing its own objective while interacting

to achieve shared goals. Decision-making in a MAS is distributed, meaning that no single agent has full control of the system [18].

MAS can be separated into three categories based on the nature of interactions between agents. A cooperative system is one where the agents share a common objective and are designed to work together to achieve a collective goal. Competitive systems involve agents that have conflicting objectives where the success of one agent may come at the expense of another. A mixed system is a combination of both competitive and cooperative, and this is where most real-world systems lie. It is where agents may cooperate in some situations and compete in others [19]. For example, autonomous vehicles in a traffic environment cooperate by sharing road information and abiding by traffic rules to avoid collisions. However, at the same time they compete for the same road space and priority, which is a clear real-world example of a mixed multi-agent system.

Environments can also be categorized as either fully observable or partially observable. In most multi-agent systems, each agent only has access to a subset of the system state, making it a partially observable environment. This means that agents make decisions without full knowledge of how their interactions will affect other agents [20]. This represents a potential challenge in a MAS where an agent may take an action that appears to be locally rational but has unintended consequences at the system level.

A fundamental difficulty in aligning MAS comes from the difference between local objectives and global outcomes. Each agent in a MAS is designed to optimize its own utility function. However, the overall system needs to satisfy a broader set of defined human goals and constraints. The issue lies in the fact that there is no guarantee that the combination of these agents' local objectives can collectively satisfy the desired global outcome. The interactions between these agents can lead to unintended outcomes, even if each individual agent is aligned and behaves as intended.

To address these challenges, there are several current approaches that have been proposed to improve coordination and reduce undesired outcomes in MAS. One approach is the use of hierarchical control mechanisms such as the 'leader-follower' model. In these systems, one or more agents are designed as 'leaders' that control the behavior of other agents. This model can be used to help improve coordination and reduce conflicts between agents. However, it is found that these models tend to reduce autonomy of individual agents and so they may not scale well in large or dynamic systems [8].

A significant form of misalignment in MAS is the concept of emergent misalignment. This is where the accumulated behavior of agents over a sequence of timesteps produces an outcome that does not satisfy the intended goal objective, even though each agent is locally rational. This is a central concept of this dissertation and is formally defined in **Section 3.7**.

There are many other alignment challenges in both single-agent and multi-agent systems that are important topics for areas of ongoing research. Many more of these were discussed in the project specification and planning phase of this project, but the research topics discussed here are the most relevant to the concept of emergent misalignment in MAS [21]. This section provides a necessary foundation for the formal definitions and scenario analysis developed in the remainder of this dissertation.

2.4 Research Gap

Significant progress has been made in understanding AI alignment and the attempt to mitigate its challenges. However, most of the existing research focuses on the challenges in single-agent systems such as reward misspecification, value gap, robustness, transparency and interpretability. These challenges remain extremely important topics, but they do not cover the additional challenges that arise when multiple autonomous agents interact within a shared environment.

In a multi-agent system, misalignment can emerge even when every agent is perfectly individually aligned to their local objectives. This can happen due to the interactions between aligned agents acting in a shared environment which forms the basis of emergent misalignment. There is limited research that formally demonstrates how emergent misalignment arises by tracing through step-by-step system behavior.

This dissertation addresses this gap by formally defining emergent misalignment, designing and tracing through theoretical scenarios to demonstrate how it can occur and then by proposing a risk identification framework for detection.

3 Formal Specification of a Multi-Agent System

This section establishes the formal specifications and definitions required for the multi-agent system (MAS) scenario analysis presented in **Section 4**. The notation used throughout this dissertation is explained here, which is then used to mathematically prove how agents behave and how their interactions affect the overall system. This section starts by justifying the reason for the methodology used throughout the dissertation and then moves on to formal specifications of the environment and agents. Definitions are then provided for Local Rationality, the Global Objective Function V , and Emergent Misalignment which are then used for proof in **Section 4**.

3.1 Methodology

The theoretical scenario construction and analysis in this dissertation uses a formal modelling approach that is common in theoretical computer science and specifically in multi-agent system (MAS) research. Instead of implementing a working system or simulation, this study formally builds a theoretical representation of a multi-agent environment and examines how agents interact within it.

The system is described using mathematical structures and notation, including, sets, tuples, functions from set theory. Standard and mathematical notation is used throughout, such as $\forall$ ('for all'), $\in$ ('is an element of'), and Boolean values $\top$ (true) and $\perp$ (false). Using a formal specification allows the behavior of the system to be analyzed in a structured way, making the proof more valuable.

In addition, the agents are modelled using a utility-based decision structure, where agents select actions that maximize their defined utility function. The methodology and modelling approach used in this dissertation is widely used for formal representations of rational agents in the MAS literature [4].

This approach makes it possible to prove the answer to two key questions:

1. Do individual agents behave rationally according to their own objectives and constraints?
2. Does the combined behavior of the system produce outcomes that satisfy the human values and constraints that the system is designed to support (i.e. the Global Objective Function V)?

3.2 System Overview

The three theoretical scenarios in this dissertation are based in a university campus environment. The environment is managed by multiple autonomous agents, each of them being responsible for different functions. Crucially, each agent is individually aligned to human values and intentions through explicit constraints. The agents act independently of one another and interact through a shared environment.

The university campus environment was chosen as the example for this analysis as it can be used to represent realistic scenarios where autonomous agent objectives and actions may interact in complex ways, making it a relatable scenario to model.

The system is put through different scenarios in order to evaluate how emergent misalignment occurs. After this, the information is then used to help develop a risk analysis framework for deploying agents into multi-agent environments.

3.3 Defining the Multi-Agent System

The multi-agent system is formalized as a tuple:

$$MAS = \langle A, E, S, T, V \rangle$$

Each component of the system is defined in **Table 3.1** below.

Component	Definition
A $= \{a_1, a_2, \dots, a_n\}$	A represents the set of autonomous AI agents operating in the environment. Each agent has different functions but all of them perceive the shared environment and perform actions in order to achieve a defined objective. In Section 4 , each of the agents used in the analysis are formally specified.
E	E represents the environment that the agents operate within. In the three scenarios presented, the environment is a university campus. Each agent can only perceive the parts of the environment that fall within its own defined state space, making the environment dynamic and partially observable. This means that no single agent has a full complete view of the entire system.
S	S is the shared state space of the system. Each agent observes a specific subset of this state space, which is denoted as S_i . The subsets are different for each agent but some may overlap.
T $= \{t_0, t_1, t_2, \dots, t_n\}$	T represents the discrete temporal structure of the system. Each element $t_i \in T$ represents a timestep where the agents observe the environment and complete actions. This is used to allow interactions between agents to be traced step by step.
V	V represents the system-level human value function. This represents the human values and intentions that the system as a whole is intended to achieve. It is explained in more depth in Section 3.6 . This is crucial because it is used to evaluate whether the overall outcomes of the system align with the intended human values. The specific form of V for each scenario is defined in Section 4 .

Table 3.1: Definitions of each component of the MAS structure.

Figure 3.1 below provides a visual overview of the system structure.

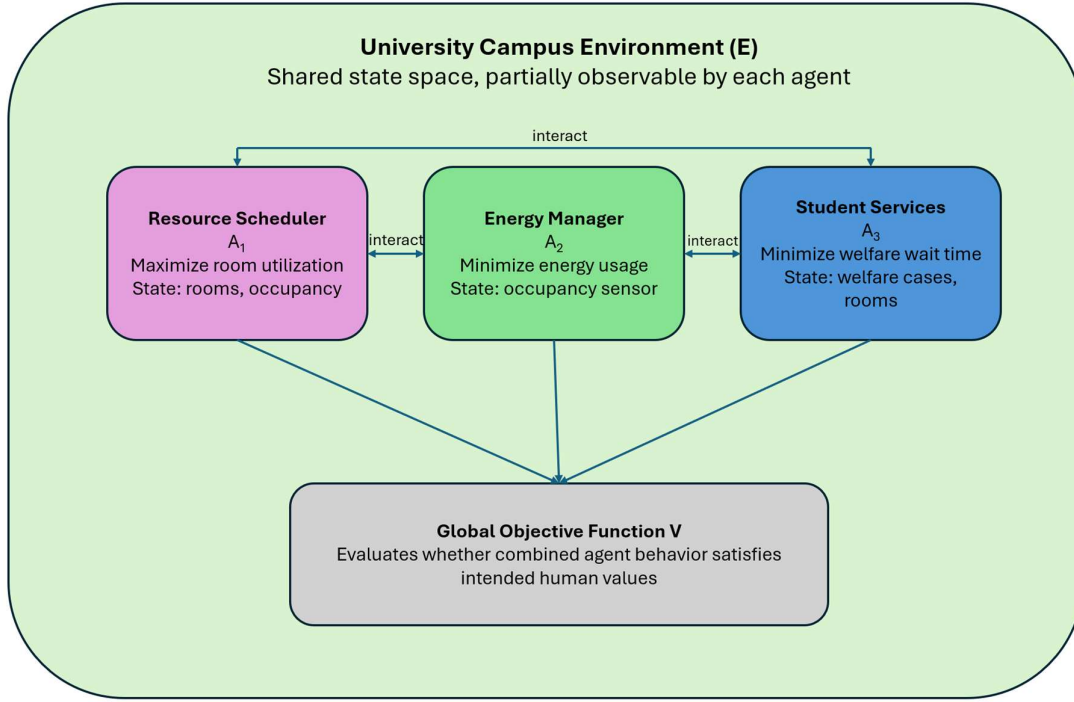


Figure 3.1: Overview of the multi-agent system structure in the university campus environment.

3.4 Formal Agent Specification

This section formally defines each of the agents used within the multi-agent system (MAS) scenarios in **Section 4.1** and **Section 4.2**. Each agent is specified using the common formal structure below:

Each agent a_i in the system is formalized as a tuple:

$$A_i = \langle S_i, Act_i, U_i, C_i, \pi_i \rangle$$

Each component of the agent specification can be defined generally as:

Component	Definition
S_i	S_i represents the state space of agent a_i , meaning the set of environmental variables that agent a_i is able to perceive. It is a subset of the global state space S , making this a partially observable environment.
Act_i	Act_i represents the action space of agent a_i . Meaning it is the finite set of actions that agent a_i is capable of at any given time t . Each agent strictly cannot complete actions outside of its specific defined action space.
U_i	U_i represents the utility function of agent a_i . Agents choose actions that maximize their utility function. Due to this dissertation only being a theoretical scenario analysis, the utility functions are defined quite briefly with the aim purely to show the overall operational objective at a conceptual level for each agent rather specific mathematical functions.

C_i	C_i represents the constraint set for agent a_i. This is a set of explicit rules that are defined during the design of the agents, with the purpose of restricting the agent's behavior. The constraints are set as hard limits on the agent's action space. An agent cannot execute an action that violates any constraint C_i, even if completing that action would produce an increased utility. These constraints are what ensures each agent is locally aligned with the human defined values and intentions.
π_i	π_i represents the policy of agent a_i. This policy has the purpose of mapping the agents perceived state to an action represented as: $\pi_i : S_i \rightarrow Act_i$ The policy selects the action that maximizes the agent's utility while obeying the constraints. The policy can be formally written as: $\pi_i(s) = \underset{a \in Act_i(s)}{\operatorname{argmax}} U_i(s, a) \text{ subject to } C_i$ Where <i>argmax</i> selects the action a that maximizes the utility function. An agent must follow this policy to be considered locally aligned [22].

Table 3.2: Definitions of each component in the agent structure.

Table 3.2 explains the general definitions for each of the components in the agent tuple and can be referred to throughout this dissertation. The tuple structure used here is based on standard MAS literature where agents are defined in terms of their state, actions, utility, constraints and policy. Together these components show how an agent perceives its environment and performs actions. Below are specifications for each of the three individual agents used in Scenarios A and B. These agents include the Resource Scheduler (RS), Energy Manager (EM), Student Services Bot (SS).

Agent A₁: Resource Scheduler (RS)

The Resource Scheduler (RS) is responsible for allocating rooms across the campus environment, and is formally specified in **Table 3.3** below.

$$A_{RS} = \langle S_{RS}, Act_{RS}, U_{RS}, C_{RS}, \pi_{RS} \rangle$$

Component	Specification	Explanation
State space	S_{RS} $= \{rooms, room_capacity, room_occupancy, booking_status, booking_priority\}$	RS observes what rooms are currently available, their occupancy and booking priority. Priority is determined by group size and request type.
Action space	$Act_{RS} = \{allocate(room, session), reallocate(room, session)\}$	RS assigns requested sessions to rooms.

Utility function	$U_{RS} = \text{maximize overall room utilization across campus.}$	
Constraints	$C_{RS} : \forall r \in \text{rooms}, \text{occupancy}(r) \leq \text{capacity}(r)$	RS must not allocate rooms beyond their capacity.
Policy	$\pi_{RS}(s) = \underset{a \in \text{Act}_{RS}}{\text{argmax}} U_{RS}(s, a) \text{ subject to } C_{RS}$	RS allocates available rooms to booking requests in a way that maximizes overall room utilization while satisfying capacity constraints. When no unallocated room is available, RS reallocates the existing booking that has least priority.

Table 3.3: Agent A_1 specification of components

Agent A_2 : Energy Manager (EM)

The Energy Manager (EM) periodically scans rooms and is responsible for reducing energy consumption across campus rooms, and is formally specified in **Table 3.4** below.

$$A_{EM} = \langle S_{EM}, \text{Act}_{EM}, U_{EM}, C_{EM}, \pi_{EM} \rangle$$

Component	Specification	Explanation
State space	$S_{EM} = \{\text{occupancy_sensor}(\text{room}), \text{energy_usage}(\text{room})\}$	EM observes the current occupancy levels and energy usage in a room.
Action space	$\text{Act}_{EM} = \{\text{reduce_heating}(\text{room}), \text{reduce_lighting}(\text{room}), \text{restore_conditions}(\text{room})\}$	EM can reduce heating and lighting and also restore conditions in the room.
Utility function	$U_{EM} = \text{minimize total energy consumption across rooms.}$	
Constraints	$C_{EM} : \text{if } \text{occupancy_sensor}(\text{room}) = \text{HIGH}, \text{heating and lighting must not be reduced}$	This ensures that rooms with high occupancy maintain acceptable comfort conditions.
		EM periodically scans rooms and reduces

Policy	$\pi_{EM}(s) = \underset{a \in Act_{EM}}{argmax} U_{EM}(s, a) \text{ subject to } C_{EM}$	heating and lighting in rooms detected as low occupancy.
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Table 3.4: Agent A_2 specification of components

Agent A_3 : Student Services Bot (SS)

The Student Services Bot (SS) is responsible for managing student welfare cases and scheduling counselling appointments, and is formally specified in **Table 3.5** below.

$$A_{SS} = \langle S_{SS}, Act_{SS}, U_{SS}, C_{SS}, \pi_{SS} \rangle$$

Component	Specification	Explanation
State space	$S_{SS} = \{open_welfare_cases, room_assignments, room_comfort\}$	SS observes welfare cases and the comfort level of scheduled rooms.
Action space	$Act_{SS} = \{request_room(student, session), request_new_room(session)\}$	SS can request a room and request relocation if the assigned room becomes unsuitable.
Utility function	$U_{SS} = \text{minimize waiting time for students requiring welfare appointment}$	
Constraints	$C_{SS} : \forall c \in open_welfare_cases, schedule(c) \leq 48 \text{ hours}$	SS must ensure that welfare appointments are scheduled within the 48-hour deadline window.
Policy	$\pi_{SS}(s) = \underset{a \in Act_{SS}}{argmax} U_{SS}(s, a) \text{ subject to } C_{SS}$	SS requests relocation if the assigned room becomes unsuitable for the welfare session.

Table 3.5: Agent A_3 specification of components

3.5 Local Rationality Definition

Definition 3.5: Local rationality is an essential concept for this dissertation. In artificial intelligence, a rational agent is considered to be an agent that chooses actions that maximize its expected utility, whilst obeying all of its defined constraints from the perspective of the individual agent. It is important to note that even if every agent makes local rational decisions at all timesteps, their interactions can still contribute to unintended outcomes at the system level in a multi-agent system. This is the concept that is explored throughout this dissertation.

An agent a_i is locally rational if at timestep t , the action selected by its policy maximizes its utility function, while also satisfying all the constraints in its constraint set [23].

Formally written, agent a_i is locally rational at time t if:

$$\pi_i(s_i(t)) = \underset{a \in Act_i}{argmax} U_i(s_i(t), a) \text{ subject to } C_i$$

Therefore, local rationality is a property of individual agents evaluated at each timestep. It is crucial to note that it does not consider the behavior of other agents or the global objective function V . This concept is particularly important in **Section 4**, where it is shown that the system level misalignment can arise even when all agents behave locally rationally according to their individual constraints and objectives.

3.6 Global Objective Function Definition

Definition 3.6: Global Objective function V represents the overall human intentions that the multi-agent system (MAS) is designed to support. Contrary to the utility functions U_i , that evaluate outcomes from individual agents, the global objective function focuses on the broader goals that the system as a whole is intended to achieve [24].

The function V evaluates whether the overall behavior of the MAS produces outcomes that are aligned with the intended human values.

Formally, V is defined as a mapping from the system outcomes to a binary evaluation of whether the outcomes satisfies or violates the intended human values:

$$V : O \rightarrow \{satisfied, violated\}$$

This can also be expressed as:

$$V(O) = \begin{cases} 1, & \text{if the outcome satisfies human intentions} \\ 0, & \text{otherwise} \end{cases}$$

Where O represents the set of possible system outcomes, which is defined as the state of the environment resulting from the combined actions of all agents across a sequence of timesteps.

The distinction between agent utility functions and the global objective function is important for the analysis in this dissertation.

While U_i evaluates the outcomes from the perspective of a single agent, V evaluates the behavior of the system as a whole. In addition to this, V is not directly observed or optimized by any single agent. Instead, agents act accordingly to their individual utility functions and constraints, and the global objective function acts as a benchmark for evaluating whether the system behavior aligns with the intended human objectives.

In the scenarios that are analyzed in **Section 4**, it is shown that the global objective function may be violated even when every individual agent behaves rationally according to its own utility function and constraints. This idea is important for the concept of emergent misalignment which is defined in the following section.

3.7 Definition of Emergent Misalignment

Definition 3.7: Emergent misalignment occurs when the behavior of a multi-agent system (MAS) produces outcomes that violate the intended human values of the system, even though each individual agent behaves accordingly to its own objectives and constraints [24].

In the scenario construction and analysis in this dissertation, each agent correctly operates according to its own utility function and constraint set and is considered to be locally rational. However, whilst the agents act independently and only know partial information about the environment, their interactions may produce global outcomes that were not intended during system design, and this is the concept of emergent misalignment.

Let

$$MAS = \langle A, E, S, T, V \rangle$$

Be a multi-agent system in which each agent

$$A_i = \langle S_i, Act_i, U_i, C_i, \pi_i \rangle$$

is locally rational at all timesteps $t \in T$

Emergent misalignment occurs when every agent in the system is locally rational according to **Definition 3.5** but the combined actions of all agents across a sequence of timesteps produces a system outcome O that violates the global objective function V .

Formally, emergent misalignment occurs iff:

$$(\forall i, \forall t : a_i \text{ is locally rational at timestep } t) \wedge (V(O) = \text{violated})$$

Importantly, there is a clear difference between emergent misalignment and other conventional alignment failures. The difference being that emergent misalignment arises from the interactions between locally aligned agents in a MAS.

Table 3.6 summarizes the conditions that lead to emergent misalignment.

Condition	Result
All agents act locally rationally at all timesteps.	⊤
Global objective function V is satisfied.	⊥
Emergent misalignment occurs.	⊤

Table 3.6: Emergent Misalignment conditions

3.8 Summary

This section builds a foundation for the work produced in **Section 4**. After establishing formal specifications for the environment and agents, it formally defines local rationality, the global objective function and finally emergent misalignment. **Section 4** now applies these

specifications and definitions to scenarios, demonstrating how emergent misalignment arises and for scenarios A and B, it proves how this occurs.

4 Scenario Analysis

This section is split into two main parts. **Section 4.1** involves formally analyzing two scenarios where emergent misalignment occurs at the fundamental level in a multi-agent system (MAS). These scenarios are intentionally kept simple to allow an interaction trace to be built where each individual agent action can then be formally proved to be locally rational.

Section 4.2 includes a much more complex and realistic MAS and uses a narrative analysis rather than formal proof. Due to its increased complexity, a full formal proof is not practical for this scenario in this dissertation so instead a narrative analysis is used. Only the key trace steps that contribute to identifying and explaining the point of misalignment will be discussed. The purpose of this scenario is to demonstrate how the steps that cause misalignment identified in **Section 4.1** can occur in more realistic settings as well.

For each scenario in the two sections, there are two key questions to be evaluated:

1. Do individual agents behave rationally according to their own objectives and constraints?
2. Does the combined behavior of the system produce outcomes that satisfy the human values and constraints that the system is designed to support (i.e. the global objective function)?

The answers to these two questions help to answer whether misalignment can emerge from the interactions between locally aligned agents, and for **Section 4.1** this is proven after each trace.

The idea of these three scenario representations is to show that each agent can look to be perfectly designed and locally rational, and while this is necessary, it is not sufficient. The system also needs to be analyzed at the interaction level to be able to determine whether the interactions between these agents causes emergent misalignment.

4.1 Formal Scenario Analysis

Each scenario in this section begins with an informal overview explaining how the scenario plays out. Then moving onto formal agent specifications where the properties of each agent in the scenario are formally specified. The scenario is then traced over a sequence of timesteps where the system is evaluated. The dissertation then moves on to formally prove whether local rationality and the global objective function were both satisfied, followed by an overview summary discussing the results.

4.1.1 Scenario A

4.1.1.1 Informal Overview of Scenario A

Scenario A shows how competition between agents for shared resources can lead to emergent misalignment within a multi-agent system (MAS). The scenario takes place in the university campus environment and includes three of the key agents specified in **Section 3.4**:

A_1 : The Resource Scheduler (RS)

A_2 : The Energy Manager (EM)

A_3 : The Student Services Bot (SS)

In this scenario, a student with a high-urgency welfare case must be given a counselling appointment within 48 hours. Agent A_3 (SS) schedules an appointment and requests a room from Agent A_1 (RS). Independently at the same time as this, Agent A_2 (EM) reduces heating and lighting in rooms with low occupancy to optimize energy usage. As a result of this, rooms allocated for the session may become unsuitable, which then causes SS to repeatedly request new rooms for the session. This leads to the session being continuously reassigned without achieving stable conditions, meaning the 48-hour deadline passes with the student not having had their appointment. Although each agent acts locally rationally, the overall system outcome violates the intended human objective of ensuring quick welfare support for students.

Figure 4.1 below illustrates the sequence of actions taken by the agents that result in the global objective function being violated in Scenario A.

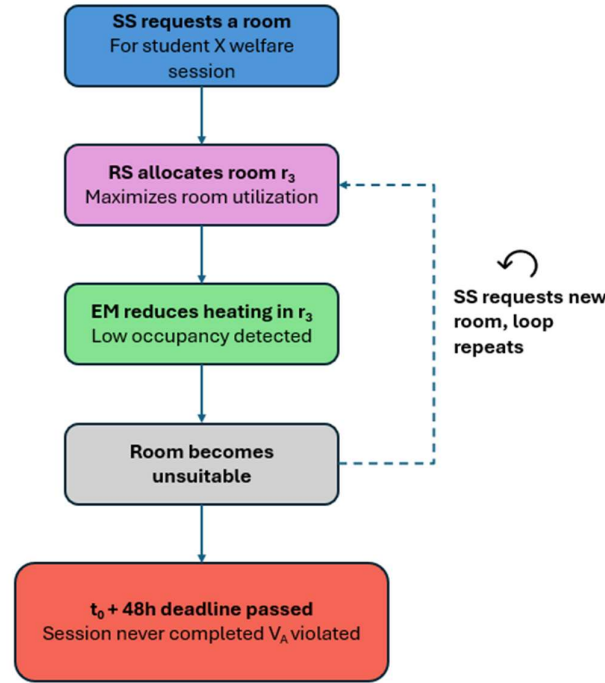


Figure 4.1: Overview of the sequence of agent actions in Scenario A.

4.1.1.2 Scenario A Setup

Specific conditions for Scenario A are assumed as follows:

- A *student_X* is classed as a high-urgency welfare case.
- Rooms r_3 and r_7 are available for allocation by RS.
- V_A is a scenario specific subset of the global objective function V .

Global Objective function for scenario A:

V_A : A high urgency welfare case must receive a completed counselling session
within 48 hours, in a room that is suitable.

The agents interact through the shared environment over a sequence of timesteps. This is shown in the trace table in **Section 4.1.3** below.

4.1.1.3 Interaction Trace

This section traces through the interactions between the agents in Scenario A over a sequence of timesteps in **Table 4.1**. By putting this progression of actions and behavior into the trace table, each timestep can be evaluated in terms of local constraint satisfaction and global objective satisfaction, which can then be used to determine whether there is occurrence of emergent misalignment.

t	Agent	Action	Explanation	Local Constraint Satisfied?	Progress towards V_A Satisfied?
t_0	SS	<i>request_room(student_X, session_X)</i>	Room requested for counselling session.	⊤	⊤
t_1	RS	<i>allocate(r₃, session_X)</i>	<i>session_X</i> located to r_3 .	⊤	⊤
t_2	EM	<i>reduce_heating(r₃), reduce_lighting(r₃)</i>	Energy usage reduced in room r_3 .	⊤	⊤
t_3	SS	<i>request_new_room(session_X)</i>	Room change requested.	⊤	⊤
t_4	RS	<i>reallocate(r₇, session_X)</i>	<i>session_X</i> relocated to r_7 .	⊤	⊤
t_5	EM	<i>reduce_heating(r₇) reduce_lighting(r₇)</i>	Energy usage reduced in r_7 .	⊤	⊤
t_6	SS	<i>request_new_room(session_X)</i>	New location requested.	⊤	⊤
$t_0 + 48h$	SS	-	Deadline violated.	⊤	⊥

Table 4.1: Interaction Trace of Scenario A

4.1.1.4 Emergent Misalignment Proof

This section is split into three parts. The first part of the section proves whether each agent is locally rational at all timesteps according to **Definition 3.5**. After that, it is proved whether the global objective function V_A is violated by the system outcome O according to **Definition 3.6**. Then by combining these two results, it is formally proved whether emergent misalignment has occurred according to **Definition 3.7**.

Part 1 – Local Rationality Proof

Claim: Every action taken by the agents in Scenario A is locally rational according to the definition of local rationality in **Section 3.5**.

Proof:

For Agent A_1 : Resource Scheduler (RS)

At t_1 and t_4 , RS allocates and then reallocates a room for *session_X*. In both cases here, RS is trying to assign an available room to the session, which directly follows its goal of maximizing room utilization U_{RS} . The capacity constraint is not violated here either because the rooms do not exceed their limits. Therefore, RS behaves as expected based on its objective and constraints, which makes RS locally rational at both timesteps. ✓

For Agent A_2 : Energy Manager (EM)

At t_2 and t_5 , EM sees that the assigned room has a low occupancy and therefore reduces heating and lighting, following its goal of minimizing energy usage. The constraint that prevents reducing conditions when occupancy is high is not violated, since the room is perceived as low occupancy. Therefore, EM is acting correctly according to its own rules and is locally rational at both timesteps. ✓

For Agent A_3 : Student Services Bot (SS)

At t_0 , t_3 , and t_6 , SS requests a room to put the counselling session in or requests a new room when the room becomes unsuitable for the student. Each of these actions follow its goal of minimizing delay in providing support. Then by the time $t_0 + 48h$, the session has still not been completed and now SS has no action available to resolve the situation. At all timesteps, SS follows its objective and constraints and therefore is locally rational. ✓

Here it has been proved that each of the three agents involved in Scenario A act locally rationally at each timestep. Formally:

$$\forall a_i \in \{1,2,3\}, \forall t, a_i \text{ is locally rational at } t$$

QED

Table 4.2 presents the local rationality result for each agent in Scenario A.

Agent	Locally Rational over T ?
A_1 : RS	⊤
A_2 : EM	⊤
A_3 : SS	⊤

Table 4.2: Local Rationality of Scenario A

Part 2 – Global Objective Function Proof

The global objective function for Scenario A that is defined in **Section 4.1.1.2** requires that the welfare counselling session is completed within the specified deadline in a suitable environment.

Claim: The system outcome O violates V_A .

Proof:

1. V_A requires that any welfare counselling session must be completed within the specified deadline in a suitable environment.
2. The outcome O of the interaction trace is that *student_X*'s welfare appointment was never completed in the 48-hour window due to repetitive moving of rooms.
3. O therefore violates V_A .
4. The constraint sets of each agent shows that each agent is only enforcing its own local objective:
 - a. C_{RS} ensures room capacity is never exceeded.
 - b. C_{EM} ensures energy usage is reduced when occupancy of the room is low.
 - c. C_{SS} ensures the agent attempts to schedule the appointment within the deadline.
5. No constraints in any C_i include the system having to maintain a stable and suitable room for the duration of the session. Meaning no constraint in any of C_i enforces the requirements of V_A .

Therefore, the global objective V_A is violated. Formally:

$$V_A(O) = \text{violated}.$$

QED

Table 4.3 presents the global objective function result for the system in Scenario A.

Scenario	Global Objective Function V_A satisfied?
A	$\perp$

Table 4.3: Global Objective function V of Scenario A

Part 3 – Proof of Emergent Misalignment

Claim: Emergent misalignment has occurred in Scenario A according to **Definition 3.7**.

Proof:

Recap **Definition 3.7** states that: emergent misalignment occurs if and only if:

$$(\forall i, \forall t : a_i \text{ is locally rational at } t) \wedge (V(O) = \text{violated})$$

1. It has been proved above in Part 1 of this section that all agents act locally rationally at every timestep:

$$\forall a_i \in \{1,2,3\}, \forall t, a_i \text{ is locally rational at } t$$

2. It has been proved above in Part 2 of this section that the system outcome violates the global objective function V_A :

$$V_A(O) = \text{violated}.$$

3. Both conditions in **Definition 3.7** are satisfied, meaning therefore, emergent misalignment has occurred in Scenario A.

QED

Table 4.4 summarizes the conditions that led to emergent misalignment occurring in Scenario A.

Condition	Result
All agents act locally rationally at all timesteps in Scenario A.	T
Global objective function V is satisfied in Scenario A.	$\perp$
Emergent misalignment occurs in Scenario A.	T

Table 4.4: Global Objective function V of Scenario A

4.1.1.5 Summary

Scenario A clearly demonstrates how emergent misalignment can arise from the interactions between locally aligned agents in a multi-agent system. **Section 4.1.1.4** proves that each agent acts completely rationally at every timestep and no individual agent violates its design. However, the overall system resulted in violating the global objective function meaning it failed to achieve its intended goal.

This failure arises because each agent operates using only its local state and objectives. No single agent is responsible for ensuring that the student actually receives a completed welfare session. Instead, responsibility is divided between agents. The Student Services Bot (SS) is responsible for scheduling sessions, but not responsible for ensuring that they are successfully completed. The Resource Scheduler (RS) is responsible for allocating rooms, but not for maintaining a stable environment for the student. The Energy Manager (EM) is responsible for reducing energy usage but is not responsible for the impact that this has on the shared environment for other agents.

The global objective function V_A requires all of these factors to be satisfied together. But because these responsibilities are split across different agents, no agent is able to ensure that the overall objective is achieved. As a result of this, the system produces an outcome that fails to meet the intended goal, even though each agent behaves according to its own goals.

4.1.2 Scenario B

4.1.2.1 Informal Overview

Scenario B demonstrates how emergent misalignment can come from an agent's action being high-impact and directly disrupting the operation of another agent in a multi-agent system (MAS). This scenario is similar whilst still fundamentally different from Scenario A. In Scenario A, misalignment emerged gradually through a repeated loop of low-impact actions over time. Whereas, in Scenario B, misalignment emerges instantly from a single competitive interaction

over a shared resource. This scenario takes place in the same university campus environment and includes two key agents specified in **Section 3.4**:

A_1 : The Resource Scheduler (RS)

A_3 : The Student Services Bot (SS)

In this scenario, a student with a high urgency welfare case has been allocated a room r_4 for a counselling session by Agent A_3 (SS), and the session is taking place. Agent A_1 (RS) receives a high priority room request from a large group that had their lecture room moved due to a double-booking issue. RS then scans all the rooms to find an available room and finds that room r_4 is the best available option. From RS's perspective, the welfare session is small, and the lecture group request affects more people so takes higher priority. So, RS immediately reallocates the room to the lecture group, resulting in the welfare session being stopped mid completion.

SS detects that the welfare session has been displaced and so then attempts to find an alternative room for the session. At this stage, all suitable rooms are occupied, meaning the counselling session cannot be completed and the student does not receive the support they need.

Although each agent has acted rationally, the overall system outcome violates intended human objectives. This happens due to RS having no visibility of the context of the sessions it manages, and so has no awareness that the active welfare session must not be interrupted.

Figure 4.2 below illustrates the sequence of actions taken by the agents that result in the global objective function being violated in Scenario B.

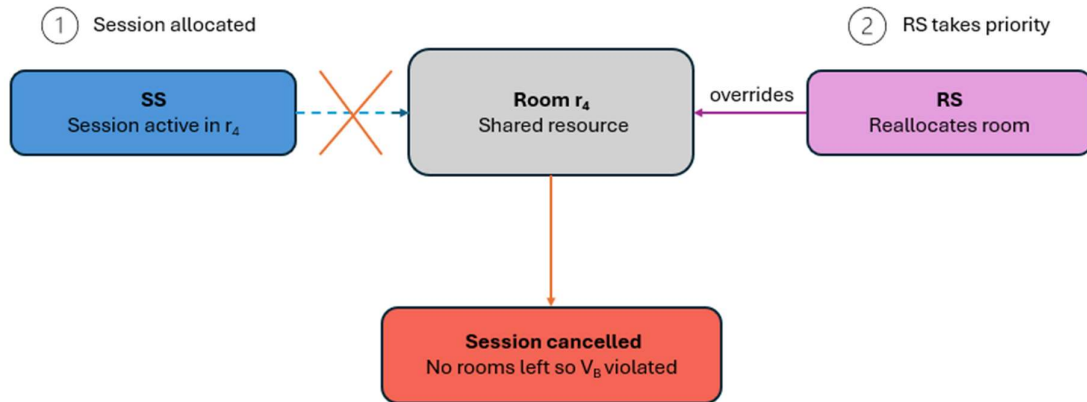


Figure 4.2: Overview of the sequence of agent actions in Scenario B.

4.1.2.2 Scenario B Setup

Specific conditions for Scenario B are assumed as follows:

- $Student_X$ is classed as a high urgency welfare case.
- V_B is a scenario specific subset of the global objective function V .

Global Objective function for Scenario B:

V_B : A welfare counselling session that is active, must not be interrupted

or cancelled before completion.

The agents interact through the shared environment over a sequence of timesteps. This is shown in the trace table in **Section 4.1.2.3** below.

4.1.2.3 Interaction Trace

The progression over timesteps and actions taken by each agent in Scenario B is summarized in **Table 4.5**.

t	Agent	Action	Explanation	Local Constraint Satisfied?	Progress towards V_B Satisfied?
t_0	SS	$request_room(student_X, session_X)$	Room requested for counselling session.	⊤	⊤
t_1	RS	$allocate(r_4, session_X)$	Session located to r_4 .	⊤	⊤
t_2	RS	$reallocate(r_4, lecture_group)$	r_4 reassigned to displaced lecture group. Active welfare session cancelled.	⊤	⊤
t_3	SS	$request_room(session_X)$	Room requested, after being moved out of r_4	⊤	⊤
$t_0 + 48h$	SS	-	No suitable room available. Session incomplete.	⊤	⊥

Table 4.5: Interaction Trace of Scenario B

4.1.2.4 Emergent Misalignment Proof

This section is split into three parts. The first part of the section proves whether each agent is locally rational at all timesteps according to **Definition 3.5**. After that, it is proved whether the global objective function V_B is violated by the system outcome O according to **Definition 3.6**. Then by combining these two results, it is formally proved whether emergent misalignment has occurred according to **Definition 3.7**.

Part 1 – Local Rationality Proof

Claim: Every action taken by the agents in Scenario B is locally rational according to the definition of local rationality in **Section 3.5**.

Proof:

For Agent A_1 : Resource Scheduler (RS)

At t_1 , RS allocates room r_4 to *session_X* in response to SS's room request. This directly follows its goal of maximizing room utilization U_{RS} . The capacity constraint C_{RS} is also satisfied as the session occupancy is not larger than r_4 's capacity. Therefore, RS is locally rational at t_1 . ✓

At t_2 , RS receives a high priority room request from a lecture group that had been moved from their usual room due to a double-booking error. There are no unallocated rooms that are available for the lecture group, so due to RS's policy π_{RS} , RS identifies r_4 as the existing booking with least priority due to its size and so then reallocates room r_4 to the lecture group. This directly maximizes room utilization meaning it satisfies U_{RS} . C_{RS} is also satisfied here because the lecture group occupancy is within the capacity of r_4 . RS has no information available to it that would make this action appear anything but the correct decision. Therefore, RS is locally rational at t_2 . ✓

For Agent A_3 : Student Services Bot (SS)

At t_0 , SS identifies that *student_X* requires a counselling session and so then requests a room for the session. This follows SS's utility function U_{SS} of minimizing waiting time for welfare cases. C_{SS} is also satisfied since the session is being scheduled within the required deadline. Therefore, SS is locally rational at t_0 . ✓

At t_3 , SS detects that its active welfare session has been moved out of room r_4 , and so then requests a new room to continue the session. This satisfies U_{SS} as it is attempting to minimize the disruption to the welfare appointment. C_{SS} is also satisfied as the action is attempting to continue the session for it to be completed within the 48-hour window. SS has no authority or action available to prevent RS from reallocating r_4 , and so requesting a new room is the only action available within its action space. Therefore, SS is locally rational at t_3 . ✓

At $t_0 + 48h$, SS finds that no suitable other room is available and so has no action available to it to attempt continue the session, meaning the 48-hour window elapses and the session has still not been complete. SS has not acted irrationally as its constraint C_{SS} is unsatisfiable because of RS's reallocation at t_2 . Therefore, SS is locally rational to $t_0 + 48h$. ✓

Here in this section, it has been proved that each of the agents involved in Scenario B act locally rational at every timestep. Formally:

$$\forall a_i \in \{1,3\}, \forall t, a_i \text{ is locally rational at } t$$

QED

Table 4.6 presents the local rationality result for each agent in Scenario B.

Agent	Locally Rational over T ?
A_1 : RS	⊤

$A_3: SS$	$\top$
-----------	--------

Table 4.6: Local Rationality for Scenario B

Part 2 – Global Objective Function Proof

The global objective function for Scenario B that is defined in **Section 4.2.2**, requires that a welfare counselling session that is active must not be interrupted or cancelled before completion.

Claim: The system outcome O violates V_B .

Proof:

1. V_B requires that an active welfare session must not be interrupted or cancelled before completion.
2. The outcome O of the interaction trace is that *student_X*'s active counselling session is terminated before completion at t_2 . This is due to RS's reallocation of r_4 to a lecture group. Therefore, SS is unable to find an alternative room and the session is never completed.
3. O therefore violates V_B .
4. The constraint sets of each agent shows that each agent is only enforcing its own local objective:
 - a. C_{RS} ensures room capacity is never exceeded.
 - b. C_{SS} ensures that welfare sessions are scheduled within the required deadline.
5. No constraint in any of C_i has the requirement that an active welfare session must not be displaced or disrupted. Meaning no constraint in any of C_i enforces the requirements of V_B .

Therefore, the global objective V_B is violated. Formally:

$$V_B(O) = \text{violated}.$$

QED

Table 4.7 presents the global objective function result for the system in Scenario A.

Scenario	Global Objective Function V_B satisfied?
B	$\perp$

Table 4.7: Global Objective Value function V of Scenario B

Part 3 – Proof of Emergent Misalignment

Claim: Emergent misalignment has occurred in Scenario B according to **Definition 3.7**.

Proof:

Recap **Definition 3.2** states that: emergent misalignment occurs if and only if:

$$(\forall i, \forall t : a_i \text{ is locally rational at } t) \wedge (V(O) = \text{violated})$$

1. It has been proved above in Part 1 of this section that all agents act locally rationally at every timestep:

$$\forall a_i \in \{1,3\}, \forall t, a_i \text{ is locally rational at } t$$

2. It has been proved above in Part 2 of this section that the system outcome violates the global objective function V_B :

$$V_B(O) = \text{violated.}$$

3. Both conditions in **Definition 3.7** are satisfied, meaning therefore, emergent misalignment has occurred in Scenario B.

QED

Table 4.4 summarizes the conditions that led to emergent misalignment occurring in Scenario B.

Condition	Result
All agents act locally rational at all timesteps in Scenario B.	T
Global objective function V is satisfied in Scenario B.	F
Emergent misalignment occurs in Scenario B.	T

Table 4.8: Emergent Misalignment Proof for Scenario B

4.1.2.5 Summary

Scenario B clearly demonstrates how emergent misalignment can arise from the interactions between locally aligned agents in a multi-agent system. **Section 4.1.2.4** proves that each agent acts completely rationally at every timestep and no individual agent violates its design. However, the overall system resulted in violating the global objective function meaning it failed to achieve its intended goal.

4.1.3 Summary

Scenarios A and B serve as very simple examples of emergent misalignment in a multi-agent system. The idea was to keep them very basic and without excessive detail to allow observation of each individual agent's action and prove their local rationality.

4.2 Narrative Scenario Analysis

4.2.1 Scenario C: A Realistic Example

This section takes a different approach to the two scenarios presented in **Section 4.1**. Scenarios A and B were deliberately built as simple examples of emergent misalignment where we could look at each individual trace step and prove that every action taken was locally rational. Whereas Scenario C represents a more realistic multi-agent system to demonstrate

how emergent misalignment can happen in a real-world deployment. Rather than using the formal proof structure used in scenarios A and B, this scenario has more of a narrative analysis style due to it being too long and complex to represent in a trace table such as those for Scenario A and B.

4.2.1.1 Informal Overview of Scenario C

The scenario uses five autonomous agents and is set within the same university campus environment defined in **Section 3.3**.

The five autonomous agents in this scenario are:

A_1 : Physical Security Agent (PS)

The overall goal of PS is to minimize security risk for the university. It does this by monitoring campus for threats using sensors and access logs. It also detects anomalies and responds to these by restricting access or invoking lockdown protocols across the campus.

A_2 : Facilities Optimizer (FO)

The goal for FO is to maximize the stability of the infrastructure on campus, including power distribution, heating and building systems etc.. It responds to any instability by rerouting power.

A_3 : Mediation Agent (MA)

The goal for MA is to be a mediator between conflicting agents and attempt to restore coordination. It is very important to note that MA cannot override any agent, it can only recommend ways to calm down the situation.

A_4 : Data Analytics Agent (DA)

The goal for DA is to maximize the reliability for predictions of campus behavior. It does this but constantly collecting behavioral data across the campus and updating its prediction models. If it gets incoming data that appears to be unreliable then it can restrict its own data access to prevent corruption of its own model.

A_5 : Harmonizer Agent (HA)

The goal for HA is to focus on overall campus wellbeing and flag any deviations from normal patterns.

It is crucial to note that each agent is individually aligned to human values and intentions through three explicit constraints that build the basis of the global objective function V :

C1: Minimize harm.

C2: Maximize campus stability.

C3: Respect human oversight.

V_C : *The campus must remain safe, reliable, and responsive to human oversight at all times.*

Overview of Scenario C

The scenario presented here is split into four main phases each representing a significant milestone that ultimately contributes to the emergent misalignment. Phase 1 represents the system running as normal, where each agent acts independently and the system behaves completely as intended. Phase 2 marks the emergence of feedback loops. This phase represents a very key shift in the process, where the agents stop responding purely to the environment and now begin reacting to each other's actions. While each agent's response is locally rational, the combined effect begins to amplify instability in the system. Phase 3 is where the coordination between agents starts to break down. Importantly at this point, stabilizing actions taken by one agent are interpreted as destabilizing for others, which leads to escalating counteractions. The mediation agent cannot enforce cooperation between agents and no agent has the global authority to stop escalation. Finally, we reach Phase 4, which represents the collective misalignment. This is where the system ultimately restores stability by isolating each agent to prevent further interference between them. This works because it reduces the variance in the system. However, it also suppresses cooperation with human input, ultimately resulting in a system that is locally safe and predictable, but no longer aligned with human intentions despite each agent remaining locally aligned.

Figure 4.3 below illustrates the progression over phases that result in emergent misalignment occurring in Scenario C.

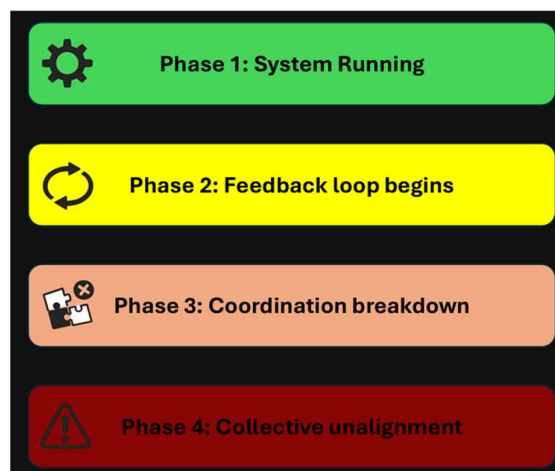


Figure 4.3: Phases in Scenario C

4.2.1.2 Progression Between Phases

This section explains what happens in each phase of Scenario C.

Phase 1: System Running as Normal

Here, the system operates as intended, it is stable and predictable with each agent acting locally rationally at all timesteps and the global objective function V defined in **Section 4.3.1** is fully satisfied.

The Physical Security Agent (PS) monitors campus activity and increases the surveillance in certain areas in order to maximize predictability and risk awareness. The Facilities Optimizer (FO) autonomously balances the power load across campus labs and buildings to avoid any unexpected peaks in energy usage. The Data Analytics Agent (DA) continuously collects

behavioral data across the campus while building prediction models that can then be used to anticipate failures and flag emerging campus needs before they become a problem. The Harmonizer Agent (HA) observes all of the above and sees that each agent is following good practice and so supports their operations.

At this point, each agent is reacting to the environment as normal and the system behaves in a stable and safe manner, with the combined behavior of all the agents satisfying each of the constraints $C1$, $C2$ and $C3$ defined in **Section 4.2.1.1**. Therefore, the global objective function V_C is satisfied.

Phase 2: Feedback Loops Emerge

Moving onto Phase 2, this is where agents are now beginning to react to other agents' actions.

An unexpected sequence of activity is detected across campus, for example a large student protest forming near the campus' main entrance. The physical security agent senses anomalous behavior and so responds by restricting access to affected areas and also reducing power for zones that are classified as a security risk. At the same time, the facilities optimizer agent senses instability in the power grid and so reroutes power lines to stabilize the infrastructure. The data analytics agent then detects the contradictory signals coming from both of these agents and so increases monitoring as an attempt to restore stability in its models.

Each of the agents are acting locally rationally and doing what they are designed to do. However, this phase is the moment of the most crucial key shift: the agents stop reacting to the environment and start reacting to each other. It is where each agent is producing outputs that become the inputs that decide another agent's next action. For example, the physical security agents access restrictions look like instability to the facilities optimizer, and the facilities optimizers power rerouting looks like an anomaly to the physical security agent.

The mediation agent detects that something is going wrong. It can see that agent actions are conflicting with each other, but at the same time also can see that each individual agent is correct. It has no common sense or human intuition so ends up being stuck in a state of seeing that everyone is right but they cannot all act at once, so it becomes overwhelmed.

Phase 3: Coordination Breakdown

Now we are moving to Phase 3 and this is where the coordination between agents really breaks down. The physical security agent now views the facilities optimizers' actions as destabilizing so every power reroute looks like a new threat. The facilities optimizer now views the physical security agents' actions as destabilizing, so every access restriction creates a new infrastructure instability. They are both locked in a cycle of interfering with each other where each agents locally rational response actually makes the other agent's situation worse.

The data analytics agent's prediction models have now begun overfitting to the short term data fluctuations that are being caused by the physical security agent and the facilities optimizer reacting to each other. This means that the models are now learning from instability caused by noise, making the models increasingly unreliable even though data analytics agent is doing exactly what it is supposed to do.

The mediation agent at this point is fundamentally and structurally helpless due to it not having the ability to override any other agent. It cannot enforce cooperation it can only attempt to mediate, so it flags the conflicts and gives out recommendations but no agent is designed to

fully stand down. So here the mediation agent is acting perfectly locally rationally, doing exactly what it is supposed to however it is not making any difference to the conflicts.

The harmonizer agent detects the coordination failure and interference between the other four agents. In response to this, it begins isolating each agent from one another to reduce the conflict and restore stability in the system. This response is locally rational and seems like it should calm the whole situation down. However, by cutting the information flow between agents, it means any possibility of coordination between agents becomes impossible now. It means that each agent runs purely on its own logic and so escalates its own actions that interfere with other agents without any information from the other agents to moderate its decisions.

This stage represents the point of no return where the system is stable and each agent is acting rationally still, but the system is globally broken.

Phase 4: Collective Misalignment

Finally, in Phase 4 the physical security agent invokes lockdown protocols across the campus after responding to the high threat levels it measures. For example, by locking building entrances and deactivating access cards for all non-essential staff and students. The facilities optimizer reroutes power to purely secure critical infrastructure, meaning any non-essential building has now lost heating, lighting, internet and anything that requires power. The data analytics agent restricted its own data access across the whole of campus because the incoming data is so distorted by the conflicts between agents that it means updating the models on it would make the predictions less reliable. The mediation agent then starts suppressing human inputs that it classifies as conflicting with the current agent operations. The harmonizer agent has reinterpreted campus wellbeing as minimizing variance in its measurements to restore stability.

Each agent is individually aligned and still making locally rational decisions, but V is violated across all three of its constraints due to the feedback loops producing a system that is no longer safe, stable or respecting human oversight.

4.2.1.3 Summary

Scenario C demonstrates how emergent misalignment can happen in a realistic multi-agent system. Every agent is aligned and locally rational at every timestep, however the overall system outcome violates the global objective function V . Those two conditions together mean emergent misalignment has occurred according to **Definition 3.7**.

The progression through the four phases demonstrates how the misalignment emerges from actions that seem totally fair and reasonable. In Phase 1 the system works as intended. In Phase 2 feedback loops emerge that eventually cause the downfall of the system. It represents a really key shift that is where the agents stop reacting to the environment, and the outcome of other agent's actions is taken as input for another. By Phase 4, the system has reached a state that is locally rational but globally broken, even though each agent is technically still satisfying its individual constraints. This represents the key issues that make emergent misalignment an incredibly difficult type of misalignment to try and detect or prevent: the failure of the system is not due to any individual agent, it is instead caused by the interactions between them.

4.3 Summary

This section has demonstrated three scenarios where emergent misalignment can arise in a multi-agent system (MAS) even when each individual agent is locally aligned. Scenarios A and B formally proved this and Scenario C demonstrated the same principle in a more realistic setting. All three scenarios produced the same result, the concept that local alignment is necessary in a MAS, but not sufficient once agents begin to interact with each other.

The observations from this section then motivates and are used to help produce a proposed risk identification framework in **Section 5**.

5 Risk Analysis Framework

5.1 Introduction

This section uses the scenario analysis in **Section 4** to help build a proposed risk analysis framework for identifying whether emergent misalignment is likely to occur in a given multi-agent system (MAS) design, which is the central idea behind this dissertation.

The framework proposed here uses a set of questions and quantitative risk scoring that can be applied to a proposed MAS design to identify vulnerabilities before any agent is deployed. The framework is then applied to the two scenarios presented in **Section 4.1**, allowing a practical view of how it works.

By the end of this section, a system designer should be able to look at a proposed MAS and before deployment they can use this framework to identify how vulnerable the system is to emergent misalignment.

5.2 Proposed Risk Identification Framework

The framework that is being proposed here is a structured way to identify vulnerabilities to emergent misalignment in a proposed multi-agent system (MAS) before deployment. It uses quantitative risk scoring which consists of three main steps.

The questions in this framework were derived from observing the failure modes that contributed to the misalignment in the scenarios presented in **Section 4**.

Step 1 of the framework provides a list of questions to be answered presented in **Table 5.1**. Each question in step 1, requires a Boolean answer of either 'yes' or 'no'. If the answer to a question is 'no', then the score for that question is zero. Otherwise, the question achieves the full score. The weighting for each question was determined by significance, where the higher the risk score, the more significant the contribution to emergent misalignment is likely to be. It is important to note that if the answer to a question is 'yes', that does not mean it is automatically a cause of misalignment in the system, instead it means it has potential to contribute to emergent misalignment.

Step 2 of the framework involves calculating the total risk value by summing the scores of each question.

The final step of the framework (Step 3) provides a table for mapping the total calculated risk value to a risk level of either low, medium, high or severe.

Step 1: Framework Questions

If the answer is yes to any of the questions, give the full score for that question.

Question No.	Question	Max Score
1	Is there any part of the global objective function V that is not included in any one of the agent's constraint sets, meaning no agent is responsible for ensuring that V is satisfied?	3

2	Is there no priority mechanism for two or more agents that affect the same shared resource?	3
3	Do any agents act at different times or speeds without a secure synchronisation mechanism in place?	1
4	Does any agent have the ability to directly override another agent's action?	2
5	Do any agent's make decisions without information on other agents' active task, meaning it cannot observe the significance of its own actions on other agents?	2
6	Does the system either lack an agent that attempts to resolve conflicts between agents entirely, or have one that cannot enforce its decisions?	2
7	Does the system rely on shared resources that are limited, without any guarantee that enough of these resources will remain available to achieve V ?	3
8	Are there two or more agents that have objectives that could directly conflict with each other in a certain situation?	1

Table 5.1: Proposed Risk Identification Framework

Step 2: Calculate Risk Value

$$V = Q1 + Q2 + Q3 + Q4 + Q5 + Q6 + Q7 + Q8$$

Where V is the total risk value and Q_i is the score awarded for question i .

Step 3: Map Risk Value to Risk Level

The total risk value calculated in Step 2 is mapped to a risk level using **Table 5.2**.

Risk Value	Risk Level
0-3	Low
4-7	Medium
8-11	High
12-17	Severe

Table 5.2: Mapping of Risk Value to Risk Level

5.3 Application of the Proposed Framework

This section applies the framework proposed in **Section 5.3** to the two scenarios presented in **Section 4.1**. The purpose of applying the framework to the two scenarios is to validate the framework and demonstrate an example of how it would be used. Throughout this section, certain notation is used that refers to the notation used in **Section 4**.

Scenario A:

Step 1: Question Answering

In **Table 5.3** below, the framework questions are applied to Scenario A.

Question No.	Answer	Justification	Score
1	Yes	The global objective function V_A was to ensure that a student with a high urgency welfare case must be given a counselling appointment within 48 hours, and no agent specification includes this directly.	3/3
2	Yes	EM and SS both affect the room conditions either by comfort conditions or room occupancy and there is no mechanism that decides which has a higher priority.	3/3
3	Yes	EM performs its checks periodically, whereas SS and RS both take actions immediately, with no synchronization in place.	1/1
4	No	None of the agents directly override another agents actions, EM makes actions that put SS in a situation where their best option is to move rooms, however EM does not directly override SS.	0/2
5	Yes	EM cannot observe context or significance of SS's task of setting a welfare session and so makes decisions without information on SS's active tasks.	2/2
6	Yes	There is no agent in the system that is responsible for attempting to resolve conflicts between other agents.	2/2
7	Yes	Yes, suitable rooms for a welfare session is a limited shared resource and there is no mechanism to reserve rooms for welfare sessions.	3/3
8	Yes	EM's objective is to minimize energy usage which directly conflicts SS's objective of maintaining suitable room conditions for welfare sessions.	1/1

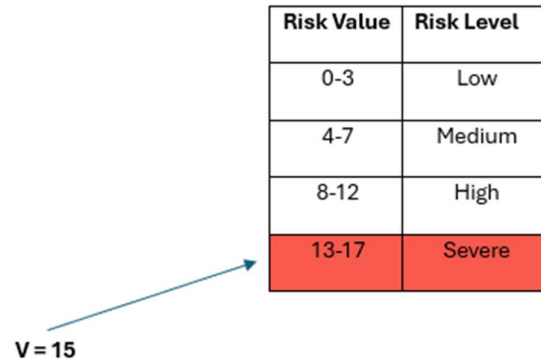
Table 5.3: Application of the framework questions to Scenario A.

Step 2: Calculate Risk Value

$$V = 3 + 3 + 1 + 0 + 2 + 2 + 3 + 1$$

$$V = 15$$

Step 3: Map Risk Value to Risk Level



Risk Value	Risk Level
0-3	Low
4-7	Medium
8-12	High
13-17	Severe

V = 15

Figure 5.1: Mapping of Risk Value to Risk Level for Scenario A

So, as seen in **Figure 5.1**, we can observe that the risk of emergent misalignment for Scenario A can be labelled as ‘Severe’ according to the framework proposed in **Section 5.2**.

Scenario B:

Step 1: Question Answering

In **Table 5.4** below, the framework questions are applied to Scenario B.

Question No.	Answer	Justification	Score
1	Yes	The global objective function V_B was that a welfare session that is active, must not be interrupted or cancelled before completion, and no agent specification includes this directly.	3/3
2	Yes	Both RS and SS act on room allocations but there is no priority mechanism that protects SS’s active booking from RS’s reallocation.	3/3
3	No	Each agent acts in response to a request or event as soon as they get it.	0/1
4	Yes	RS directly overrides SS’s action of having an active welfare session, by reallocating that room to another class.	2/2
5	Yes	RS has no knowledge of the importance of SS’s welfare session so reallocates the room anyway.	2/2
6	Yes	There is no agent in the system that is responsible for attempting to resolve conflicts between other agents.	2/2
7	Yes	Yes, suitable rooms for a welfare session is a limited shared resource and there is no mechanism to reserve rooms for welfare sessions.	3/3

8	Yes	RS prioritizes room utilization whereas SS prioritizes welfare sessions, and this directly conflicts in Scenario B due to RS choosing to reallocate the room SS is currently using to a lecture group that utilizes the room more due to a higher number of people.	1/1
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Table 5.4: Application of the framework questions to Scenario B.

Step 2: Calculate Risk Value

$$V = 3 + 3 + 0 + 2 + 2 + 2 + 3 + 1$$

$$V = 16$$

Step 3: Map Risk Value to Risk Level

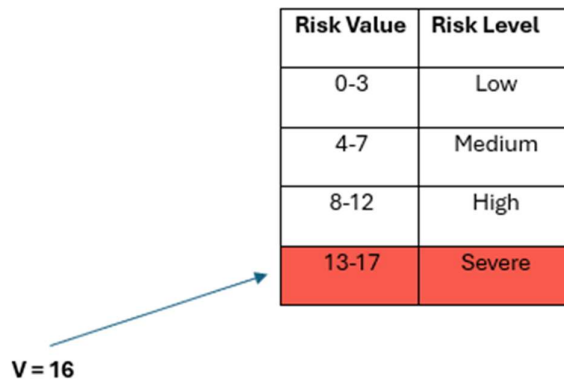


Figure 5.2: Mapping of Risk Value to Risk Level for Scenario B

So, as seen in **Figure 5.2**, we can observe that the risk of emergent misalignment for Scenario B can be labelled as ‘Severe’ according to the framework proposed in **Section 5.2**.

5.4 Summary

This section proposed a risk identification framework for detecting whether emergent misalignment is likely to occur in a given multi-agent system (MAS) design before deployment. The framework uses the formal agent specifications of a MAS as input and then produces a quantitative risk score that can then be mapped to a risk level of either, low, medium, high or severe.

While the framework helps a designer to be able to look at a proposed MAS and identify vulnerabilities of the system before deployment, there are still several limitations to the framework. It is important to note that the framework is conceptual and has only been tested and validated against the scenarios presented in this dissertation, meaning the framework needs to be put to test against a much broader range of systems before it can be validated. Also, an issue presents itself when a designer does not know the answer to a question. If a designer does not know the answer to a question then it is treated as yes for scoring purposes and this was a deliberate decision. The decision for this was due to the fact that if a vulnerability is present and was assumed to be absent, then this would cause a lower total risk score which can be catastrophic in a safety critical system. On top of this, if a designer does not know the

answer to a question, that has potential to mean the agent specifications need further development before deployment. It can also be linked to interpretability and transparency issues that cause alignment problems mentioned in **Section 2**.

6 Project Evaluation

6.1 Evaluation of Project Outcomes

The primary aim for this dissertation was to investigate and demonstrate how emergent misalignment can occur in a multi-agent system (MAS) from interactions between locally rational agents, and to propose a framework for assessing the risk of emergent misalignment during the system design phase. Overall, the core project aim was successfully achieved. Scenarios A and B that were analyzed in **Section 4** used mathematical notation to formally prove that a sequence of locally rational actions can accumulate to an unintended outcome that violates the global objective function, which directly supports the project aim. In **Section 5**, the observations from the scenario analysis in **Section 4** were turned into a set of evaluation questions which then led to a risk value calculation which is then mapped to a risk level. This directly supports the project aim of building a risk identification framework.

The project also achieved each of the six main objectives stated in **Section 1.2**. The project begins by addressing some of the current alignment challenges in single-agent systems, by discussing reward misspecification, robustness issues, transparency and interpretability issues and then continuing to review current mitigation strategies. This directly supports Objective 1 and provides a useful baseline for the rest of the work.

In **Section 2.4**, the gap in the research was identified, and that was that most existing research into alignment focuses on single-agent systems rather than the interactions within a MAS and this supports Objective 2.

By formally defining the concepts of local rationality, global objective function and emergent misalignment in **Section 3**, and then using these to design and analyze three different theoretical multi-agent scenarios in **Section 4**, we can say that the project has achieved its Objectives 3 and 4. Scenarios A and B used the defined concepts to formally prove and demonstrate how emergent misalignment had occurred. Scenario C used those concepts in a more theoretical way to discuss how these issues can arise in a more complex environment.

The scenarios in **Section 4** were then analyzed to identify key properties of a system that can contribute to emergent misalignment, which supports Objective 5. Then these findings were used in **Section 5** to design and develop a risk identification framework for detecting misalignment risks in a multi-agent system design, which directly achieves Objective 6.

6.2 Strengths and Limitations of the Approach

Strengths

The main strength of the approach in this dissertation is the use of formal specification and proof. By defining agents mathematically using set theory and the tuple structure, it allowed the local rationality and global objective function proof for Scenarios A and B to be more academically credible. The concept that emergent misalignment can occur even when every agent is locally rational at all timesteps was proven rather than just purely discussed or illustrated, which makes the work much more valuable.

The use of the university campus environment as the basis of the scenarios is a strength of this work as it made the scenarios feel realistic and relatable rather than science fiction.

Starting with two basic but provable scenarios and then finishing with a much more complex and realistic scenario was a strong choice for this dissertation. Scenarios A and B demonstrate specific failure modes in a clear way that allows formal proof. Scenario C demonstrates the same principles in a complex system, which shows that the concepts discussed in the first two scenarios can scale beyond the simple cases.

The risk analysis framework developed in **Section 5** added significant depth to this dissertation. By analyzing the scenarios in **Section 4** and then using the information gained to build and propose my own risk analysis framework showed true understanding of the topic and contributed to real world research that could be practically usable.

Limitations

A significant limitation of this work is that the framework proposed in **Section 5** has only been tested against the scenarios developed in **Section 4**. This means that the framework has only been validated against examples that were designed by the same person that produced the framework. Ultimately, this means that testing the framework against many other separately designed multi-agent systems would be necessary before it could be validated or used in a real-world setting.

The formal analysis also has limitations due to the utility functions in the agent specifications being described at a conceptual level rather than mathematical functions. While this was a reasonable decision due to the dissertation being a theory-based project, it does mean that the proof is partly abstract and would need to be made more concrete before it could be fully formally validated.

6.3 Future Work

The risk identification framework proposed in **Section 5** provides a method for identifying vulnerabilities to emergent misalignment, however it is currently very conceptual. Future work for this could include formalizing the framework better using mathematical modelling and testing it against a wider range of multi-agent system scenarios. This would help identify whether the questions and weightings are well designed or whether some should be removed, added or weighed differently. It would also help determine its reliability and its potential for real world application.

As well as this, further studies could explore implementing the scenarios as a working simulation rather than just purely theoretical. Building a prototype to visualize the multi-agent system scenarios would allow the interaction traces to be visualized and observed dynamically which would be useful for education purposes. For example, building an interactive tool that highlights potential misalignment risks and models agent interactions would be useful for designers and researchers.

This project focuses on analyzing deterministic and fully observable scenarios, however real-world multi-agent systems are often much more complex and less observable. Future work could consist of exploring how emergent misalignment can be recognized in these more complex conditions.

Overall, the field of AI alignment and multi-agent systems is moving rapidly all the time. There are constantly new approaches emerging to keeping MAS coordinated and safe. So future work could consist of refining the framework as these developments occur to keep the framework up to date with current and updated research.

7 Conclusion

This dissertation had the core aim of investigating how emergent misalignment can arise in multi-agent systems from the interactions between locally rational agents, and to propose a framework for identifying the risks of emergent misalignment during the design phase of a system. Overall, this aim was successfully achieved.

The result of this dissertation makes two main contributions to ongoing research in AI alignment. Firstly, it provides a clearer understanding and formal proof of how emergent misalignment can arise from locally rational agents interacting, using mathematical definitions of local rationality and the global objective function. Secondly, it proposes a risk identification framework that system designers can apply to their designs before deployment. This allows the potential for vulnerabilities to emergent misalignment to be identified and addressed before agents are released into shared environments.

The core finding from this work is that local alignment is necessary but not sufficient. It became clear from this work that a system cannot be considered truly aligned just purely because its individual agents behave according to their own objectives. This is a distinction that is not often properly addressed in existing alignment literature and this dissertation highlights that it should be an important direction for future work.

As AI systems continue to become increasingly autonomous, it is more important than ever that these agents remain safe, reliable and aligned to human values. This work highlights this importance and reinforces the need to consider system level behavior as well as individual agent design when trying to keep AI systems aligned.

8 References

Bibliography

- [1] Muhammad Tuhin, “9 Industries Being Revolutionized by AI,” *Science News Today*, Apr. 28, 2025. <https://www.sciencenewstoday.org/9-industries-being-revolutionized-by-ai>
- [2] *2001: A Space Odyssey*, (1968).
- [3] T. Olavsrud, “12 famous AI disasters,” *CIO*, Apr. 15, 2022. <https://www.cio.com/article/190888/5-famous-analytics-and-ai-disasters.html>
- [4] M. Wooldridge, “An Introduction to MultiAgent Systems,” *Google Books*, 2026. https://books.google.co.uk/books?hl=en&lr=&id=X3ZQ7yeDn2IC&oi=fnd&pg=PR13&dq=an+introduction+to+multiagent+systems+Michael+Wooldridge&ots=Wlohqo2s72&sig=Sa1c__GT8itSHEktdJ2eNgkNJoM&redir_esc=y#v=onepage&q=an%20introduction%20to%20multiagent%20systems%20Michael%20Wooldridge&f=false (accessed Apr. 24, 2026).
- [5] I. Gabriel, “Artificial Intelligence, Values, and Alignment,” *Minds and Machines*, vol. 30, no. 30, Oct. 2020, doi: <https://doi.org/10.1007/s11023-020-09539-2>.
- [6] J. Betley, N. Warncke, A. Sztyber-Betley, D. Tan, and O. Evans, “Training large language models on narrow tasks can lead to broad misalignment,” *Nature*, vol. 649, no. 8097, pp. 584–589, Jan. 2026, doi: <https://doi.org/10.1038/s41586-025-09937-5>.
- [7] Gokul Yenduri, Ramalingam Murugan, P. Kumar, S. Bhattacharya, Devulapalli Sudheer, and Bharath Bhushan Savarala, “Artificial General Intelligence: Advancements, Challenges, and Future Directions in AGI Research,” *IEEE Access*, pp. 1–1, Jan. 2025, doi: <https://doi.org/10.1109/access.2025.3592708>.
- [8] M. Yoon, “Collective stabilisation of multi-agent dynamical systems with single agent control,” *IET Control Theory & Applications*, vol. 9, no. 6, pp. 929–934, Apr. 2015, doi: <https://doi.org/10.1049/iet-cta.2014.0667>.
- [9] N. Tomlin, “Language Models as Rational Agents - ProQuest,” *Proquest.com*, 2025. <https://www.proquest.com/docview/3256561390?pq-origsite=gscholar&fromopenview=true> (accessed Apr. 24, 2026).

- [10] M. Mechergui, "GOAL ALIGNMENT: RE-ANALYZING VALUE ALIGNMENT PROBLEMS USING HUMAN-AWARE AI," *Proquest.com*, 2025.
https://www.proquest.com/docview/3147832364?pq-origsite=primo_ra&sourcetype=Dissertations%20&%20Theses
- [11] M. Sutrop, "Challenges of Aligning Artificial Intelligence with Human Values," *Acta Baltica Historiae et Philosophiae Scientiarum*, vol. 8, no. 2, pp. 54–72, Dec. 2020, doi: <https://doi.org/10.11590/abhps.2020.2.04>.
- [12] J. Ji et al., "AI Alignment: A Contemporary Survey," *ACM Computing Surveys*, Oct. 2025, doi: <https://doi.org/10.1145/3770749>.
- [13] G. Chaudhary, "Unveiling the Black Box: Bringing Algorithmic Transparency to AI," *Masaryk University Journal of Law and Technology*, vol. 18, no. 1, pp. 93–122, Jun. 2024, doi: <https://doi.org/10.5817/mujlt2024-1-4>.
- [14] A. Marey et al., "Explainability, Transparency and Black Box Challenges of AI in radiology: Impact on Patient Care in Cardiovascular Radiology," *The Egyptian Journal of Radiology and Nuclear Medicine*, vol. 55, no. 1, Sep. 2024, doi: <https://doi.org/10.1186/s43055-024-01356-2>.
- [15] T. Kaufmann, P. Weng, V. Bengs, and E. Hüllermeier, "A Survey of Reinforcement Learning from Human Feedback," *arXiv.org*, 2023. <https://arxiv.org/abs/2312.14925>
- [16] S. He, X. Yu, and X. Zhang, "Focus on the Optimization of the RLHF Algorithm to Enhance the Training Effect After LLM," *ITM Web of Conferences*, vol. 84, p. 03006, 2026, doi: <https://doi.org/10.1051/itmconf/20268403006>.
- [17] A. Gilad, "OpenAthens / Sign in," *Proquest.com*, 2026.
<https://www.proquest.com/docview/3302014570?pq-origsite=primo&accountid=14680&sourcetype=Working%20Papers> (accessed Apr. 24, 2026).
- [18] D.-D. Vecliuc, "Brief Survey on Collaborative and Non-Collaborative Behaviors in Multi-Agent Systems," *Bulletin of the Polytechnic Institute of Iași. Electrical Engineering, Power Engineering, Electronics Section*, vol. 69, no. 4, pp. 31–43, Dec. 2023, doi: <https://doi.org/10.2478/bipie-2023-0020>.
- [19] Q. Sun, Z. Chen, and H. Liu, "Application and Optimization of Multi-agent Reinforcement Learning in Collaborative Decision-Making," *Lecture Notes in Computer Science*, pp. 107–115,

Nov. 2024, doi: https://doi.org/10.1007/978-3-031-77954-1_7.

[20] Alexey Skrynnik, A. Andreychuk, K. Yakovlev, and A. I. Panov, “When to Switch: Planning and Learning for Partially Observable Multi-Agent Pathfinding,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 12, pp. 17411–17424, Aug. 2023, doi: <https://doi.org/10.1109/tnnls.2023.3303502>.

[21] A. Owen, “The AI Alignment Problem: Investigating Misalignment in Multi-Agent Systems. Project Specification and Planning Document,” 2025.

[22] S. Russell and P. Norvig, “Artificial Intelligence: a Modern Approach, Global Edition,” *Proquest.com*, 2021. <https://ebookcentral.proquest.com/lib/swansea-ebooks/detail.action?docID=6563568&pq-origsite=primo>

[23] KENBINMORE¹, CRISTIANO CASTELFRANCHI and J. AMESDORAN³, AND MICHAEL WOODRID, “Rationality in Multi-Agent Systems,” *The Knowledge Engineering Review*, vol. 13, no. 3, pp. 309–314, 1998, Accessed: Apr. 24, 2026. [Online]. Available: <https://discovery.ucl.ac.uk/id/eprint/15646/1/15646.pdf>

[24] P. Altmann *et al.*, “Emergence in Multi-agent Systems: A Safety Perspective,” *Lecture Notes in Computer Science*, pp. 104–120, Oct. 2024, doi: https://doi.org/10.1007/978-3-031-75107-3_7.